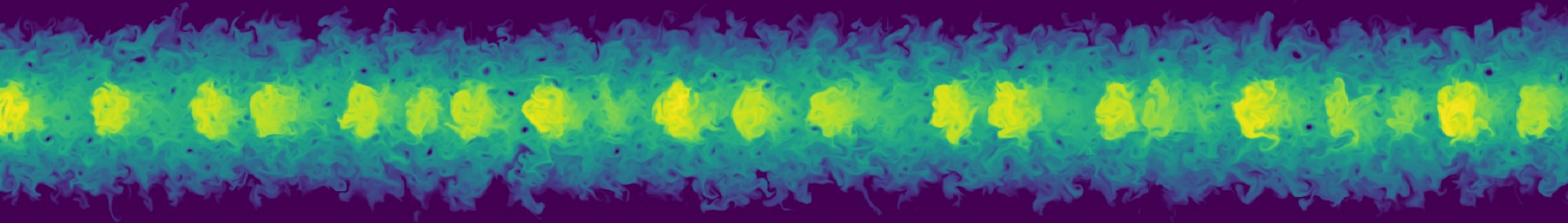


Global simulations of streaming instability



A comparison between particle and fluid approach

final year PhD student, supervised by Mario Flock (MPIA)

Motivation

- How to go from mm/cm grains to ~km planetesimals
- SI is a candidate to help dust concentrate
- Mechanism for strong clumping still unclear
 - Related to Turbulent concentration (e.g. [Yang, Zhu 21](#), [Umurhan+25](#))?
See also [Fabiola's](#), [Heloise's](#) and [Hubert's](#) talks
- How do particle / fluid simulation results differ
- Is resolution an important factor?
- What can we learn from [global \(no SB\)](#), [stratified](#) models w/ [Fluid](#), [Particle](#)-approach? (complementary work to [Stanley Baronett's](#) SI code comparison work)

What are we investigating?

D. Ostertag, M. Flock, P. Rosendahl, A. Mignone, J. B. Simon, J. Lim (in revision)

- Impact of **spatial reconstruction** schemes (PLM / PPM)
 - Impact of **Riemann solvers** (HLL / HLLC / ROE)
 - Difference PLUTO **Fluid** / PLUTO **Particles**
 - Difference between **PLUTO** Fluid / **FARGO3D**
 - **Computing performance** of different setups / codes
- Setup based on [Flock, Mignone 21](#)
2D, isothermal + axisymmetric,
St=0.01, Z=0.01,
global gas damping
- PLUTO: see [Mignone+07](#), [Mignone+19](#) (Particles), [Ziampras+25](#) (Fluid, incl. P. Sudarshan)
using also PyPLUTO for plotting: github.com/GiMattia/PyPLUTO
 - FARGO3D: Fluid, see [Benitez-Llambay, Masset 16](#), [Benitez-Llambay+19](#)

What are we investigating?

D. Ostertag, M. Flock, P. Rosendahl, A. Mignone, J. B. Simon, J. Lim (in revision)

- Impact of **spatial reconstruction** schemes (PLM / PPM)
- Impact of **Riemann solvers** (HLL / HLLC / ROE)
- Difference PLUTO **Fluid** / PLUTO **Particles**
- Difference between **PLUTO** Fluid / **FARGO3D**
- **Computing performance** of different setups / codes

Setup based on
[Flock, Mignone 21](#)

2D, isothermal +
axisymmetric,
St=0.01, Z=0.01,
global gas damping

- PLUTO: see [Mignone+07](#), [Mignone+19](#) (Particles), [Ziampras+25](#) (Fluid, incl. P. Sudarshan)
using also PyPLUTO for plotting: github.com/GiMattia/PyPLUTO

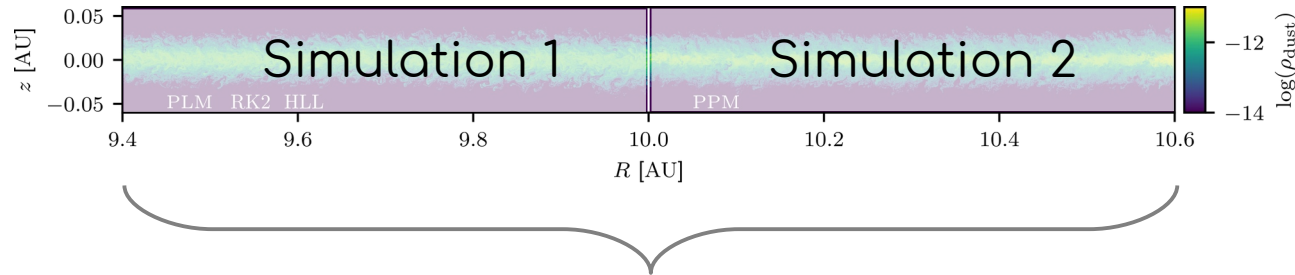
FARGO3D: Fluid, see [Benitez-Llambay, Masset 16](#), [Benitez-Llambay+19](#)

What are we investigating?

D. Ostertag, M. Flock, P. Rosendahl, A. Mignone, J. B. Simon, J. Lim (in revision)

- Impact of **spatial reconstruction** schemes (PLM / PPM)
 - Impact of **Riemann solvers** (HLL / HLLC / ROE)
 - Difference **PLUTO Fluid** / PLUTO **Particles**
 - Difference between **PLUTO Fluid** / **FARGO3D**
 - **Computing performance** of different setups / codes
- Setup based on **Flock, Mignone 21**
2D, isothermal + axisymmetric,
St=0.01, Z=0.01,
global gas damping
- PLUTO: see **Mignone+07, Mignone+19** (Particles), **Ziampras+25** (Fluid, incl. P. Sudarshan)
using also PyPLUTO for plotting: github.com/GiMattia/PyPLUTO
 - FARGO3D: Fluid, see **Benitez-Llambay, Masset 16, Benitez-Llambay+19**

Hint on Domain size



Active radial Domain size for each simulation

Vertical domain size:

reconstruction / Riemann solver: $\pm 0.1 H_g$

comparison Fluid / Particles: $\pm 0.2 H_g^*$

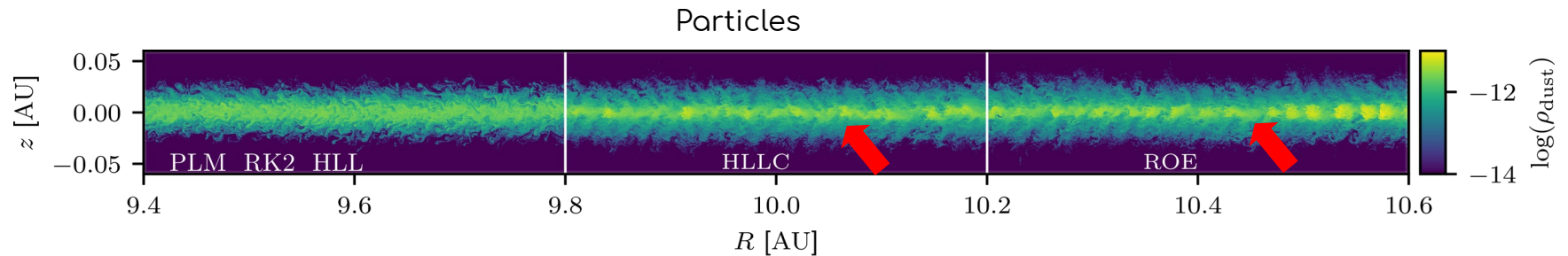
* [Li & Youdin 21](#) showed that taller boxes can lower critical Z

Influence of Gas Riemann Solvers

- Let disk evolve for 100 orbits
- Compare structure of dust density

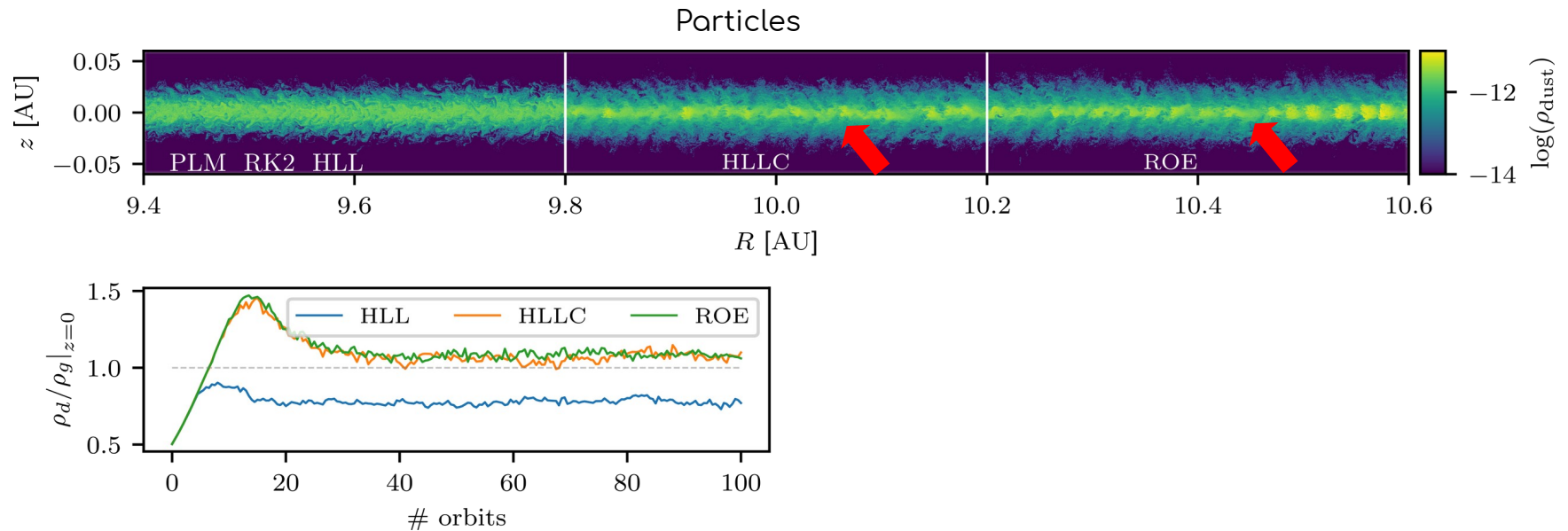
Influence of Gas Riemann Solvers

- Let disk evolve for 100 orbits
- Compare structure of dust density



Influence of Gas Riemann Solvers

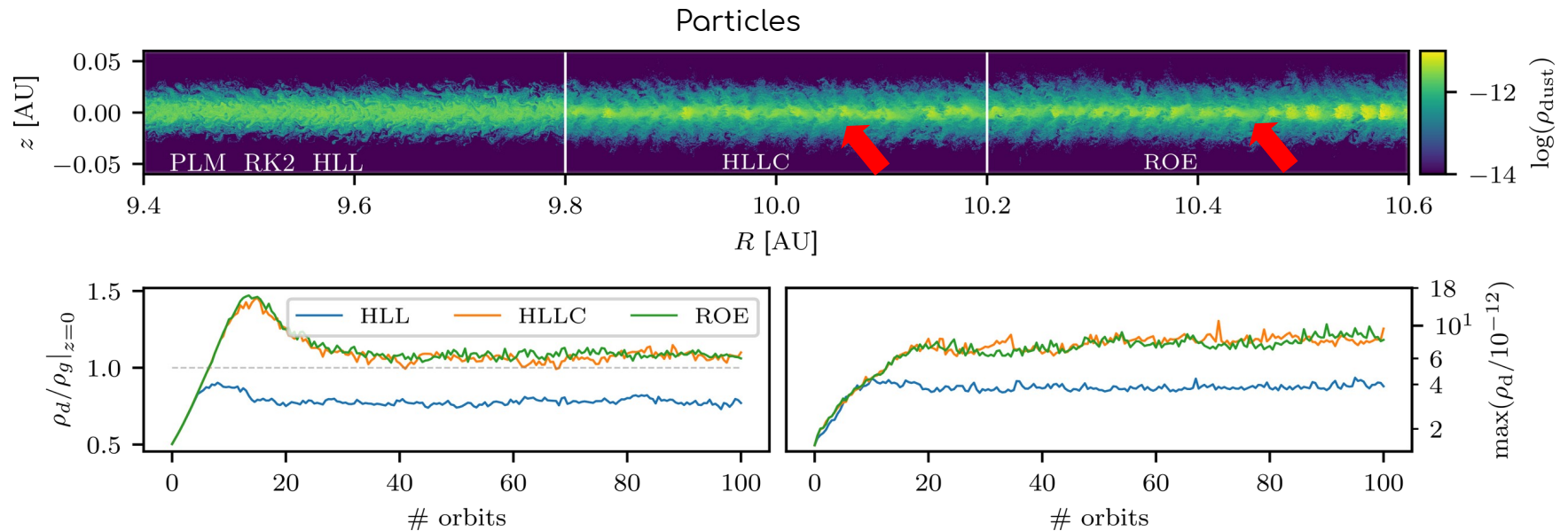
- Let disk evolve for 100 orbits
- Compare structure of dust density



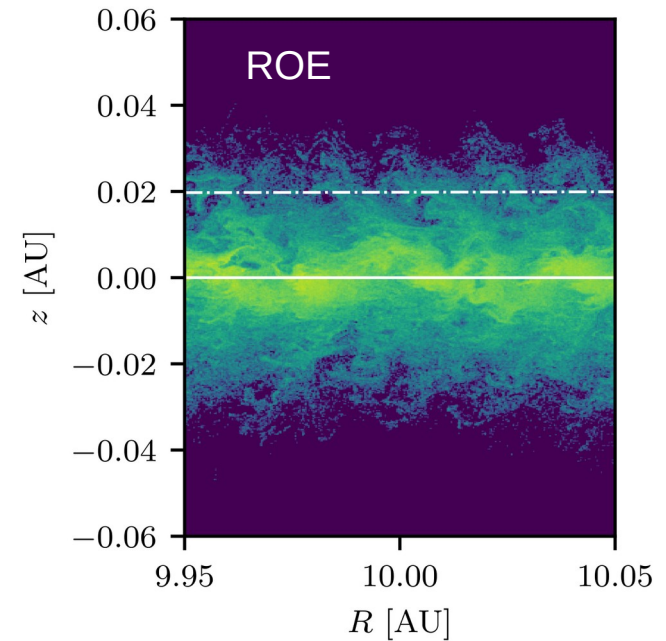
Influence of Gas Riemann Solvers

- Let disk evolve for 100 orbits
- Compare structure of dust density

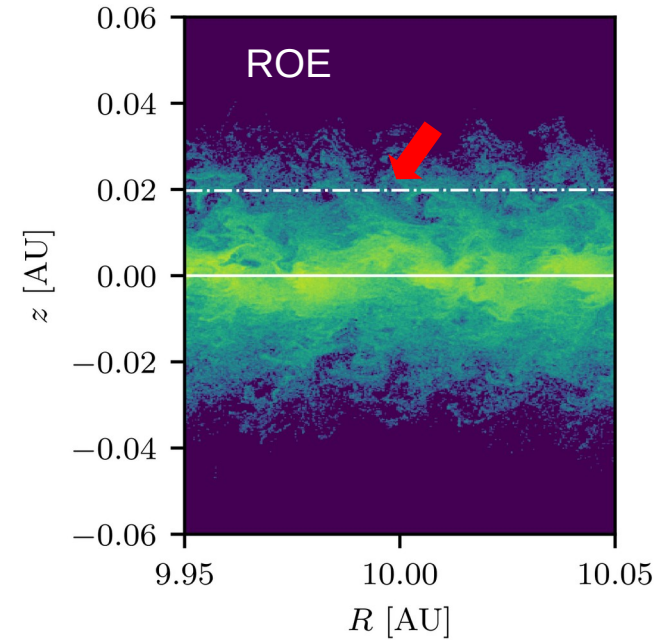
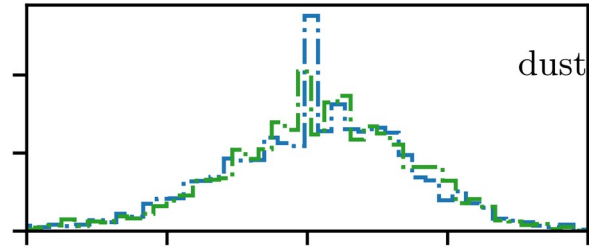
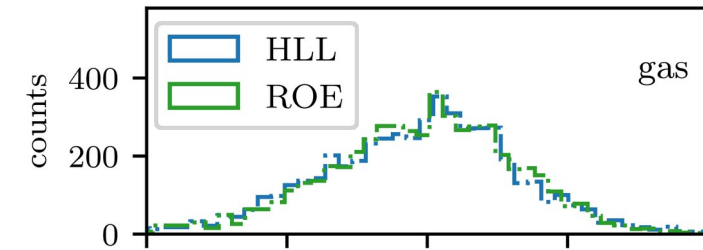
=> more accurate solver allows dust to concentrate more



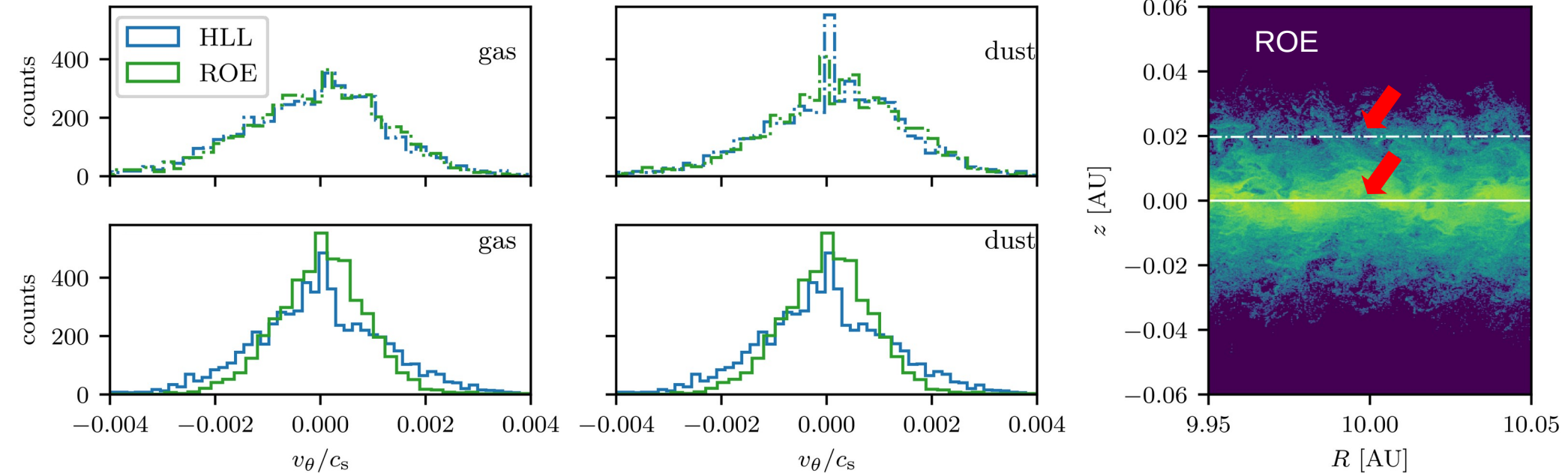
Influence of Gas Riemann Solvers



Influence of Gas Riemann Solvers

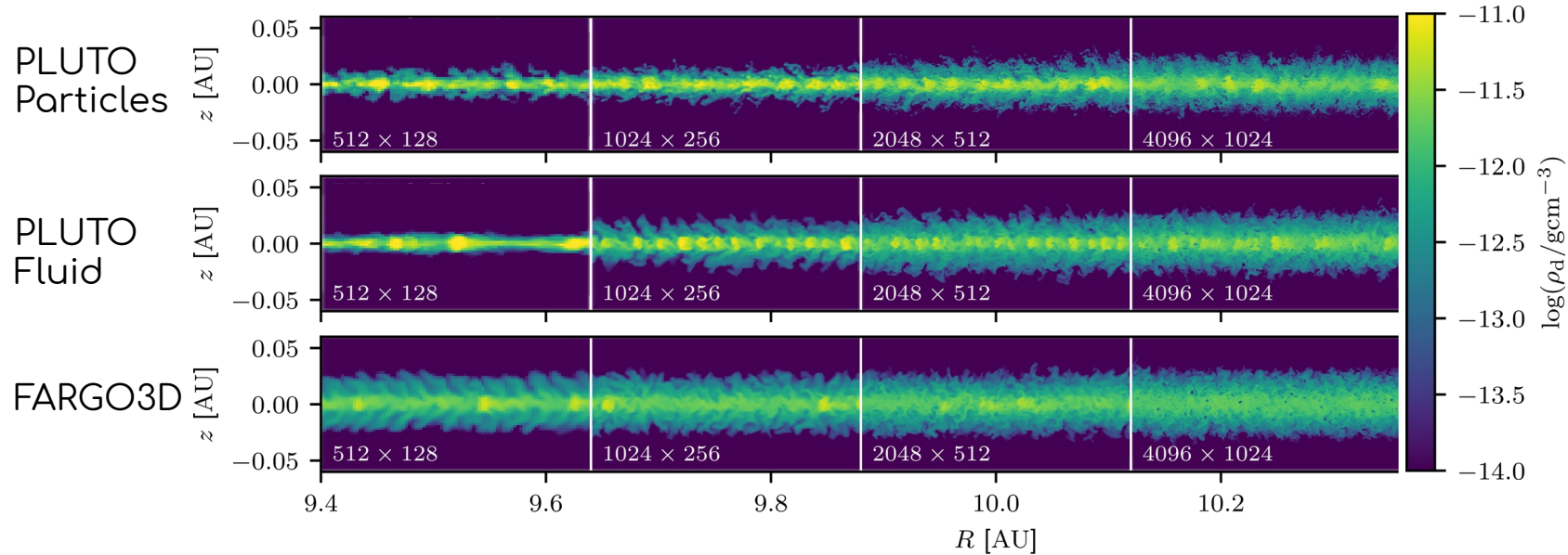


Influence of Gas Riemann Solvers

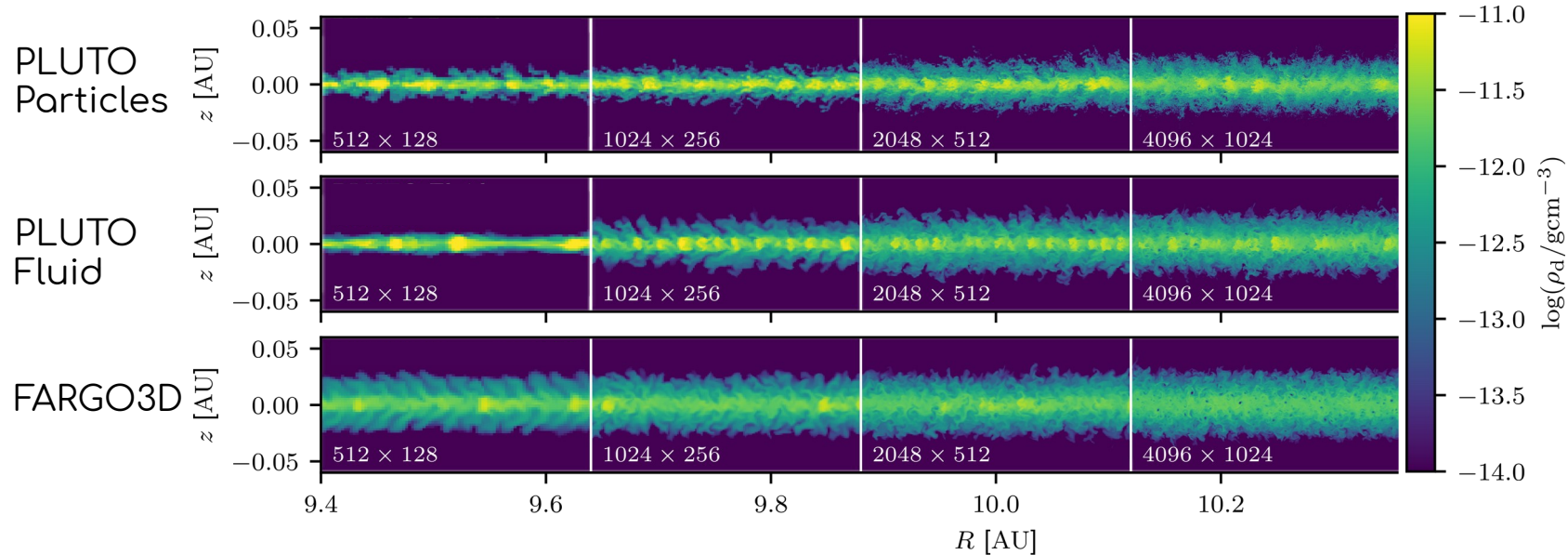


=> ROE smaller vertical velocities at midplane, compared to HLL

Fluid / Particles

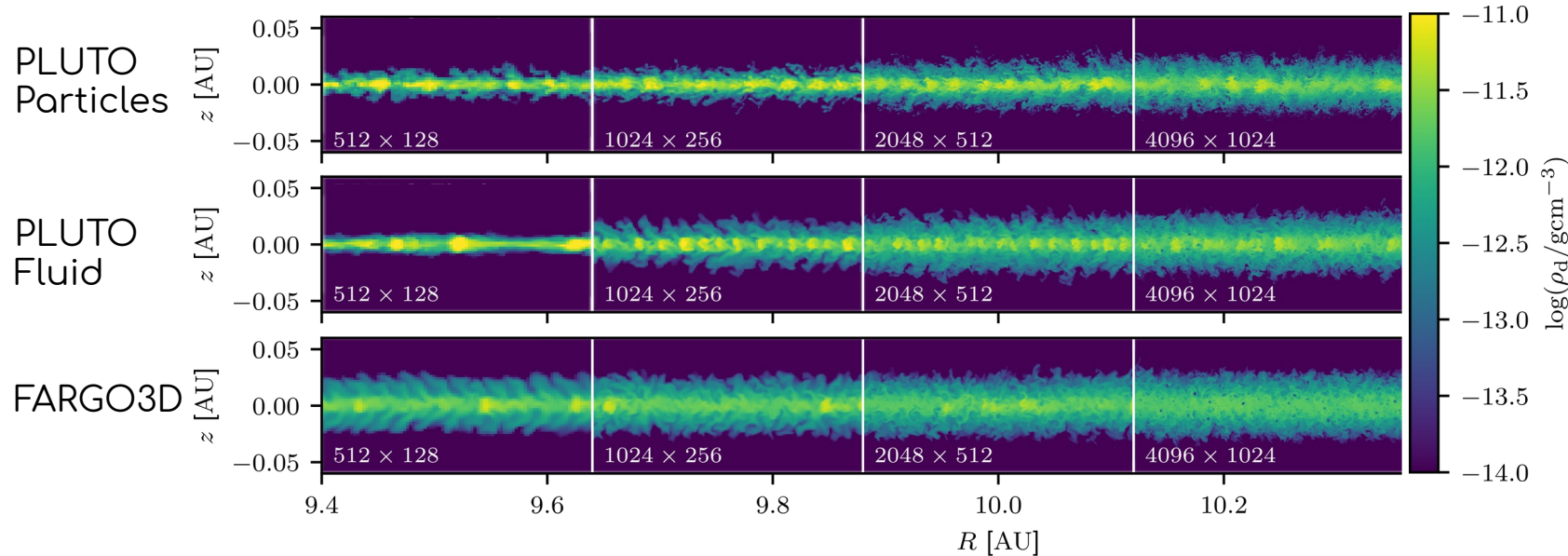


Fluid / Particles



- PLUTO Fluid / Particles **similar results**
- **Particles** still **thinner** dust **disk**
- FARGO3D **looses midplane component**, shows **first** dust depleted **voids**

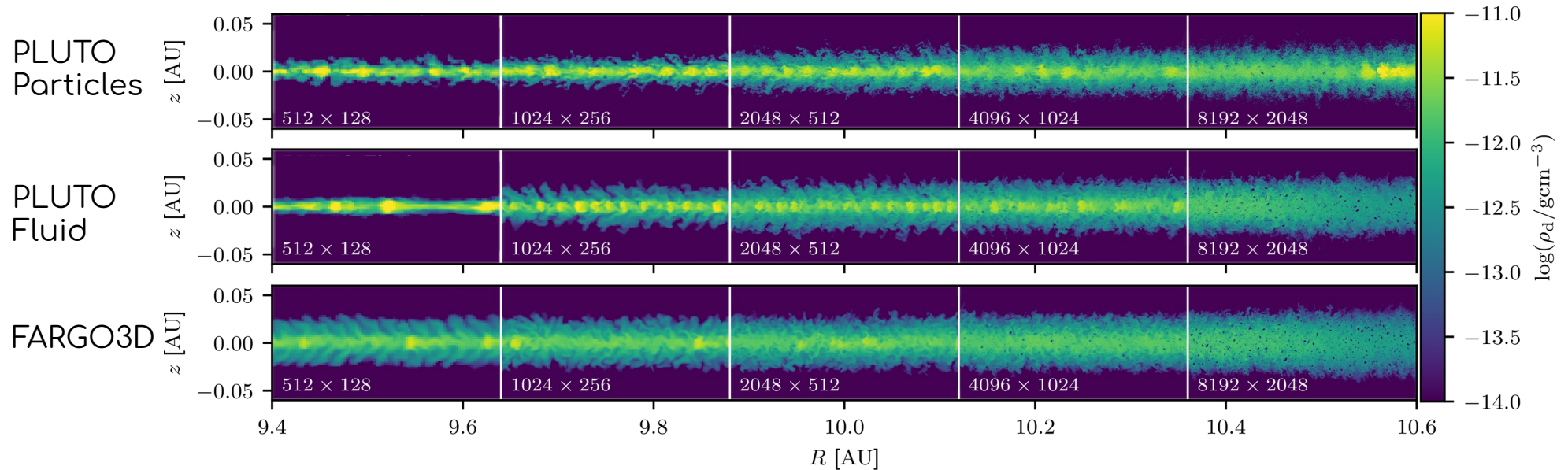
Fluid / Particles



- PLUTO Fluid / Particles **similar results**
- **Particles** still **thinner** dust **disk**
- FARGO3D **looses midplane component**, shows **first** dust depleted **voids**

Orkan, Jake: motivated for higher resolution

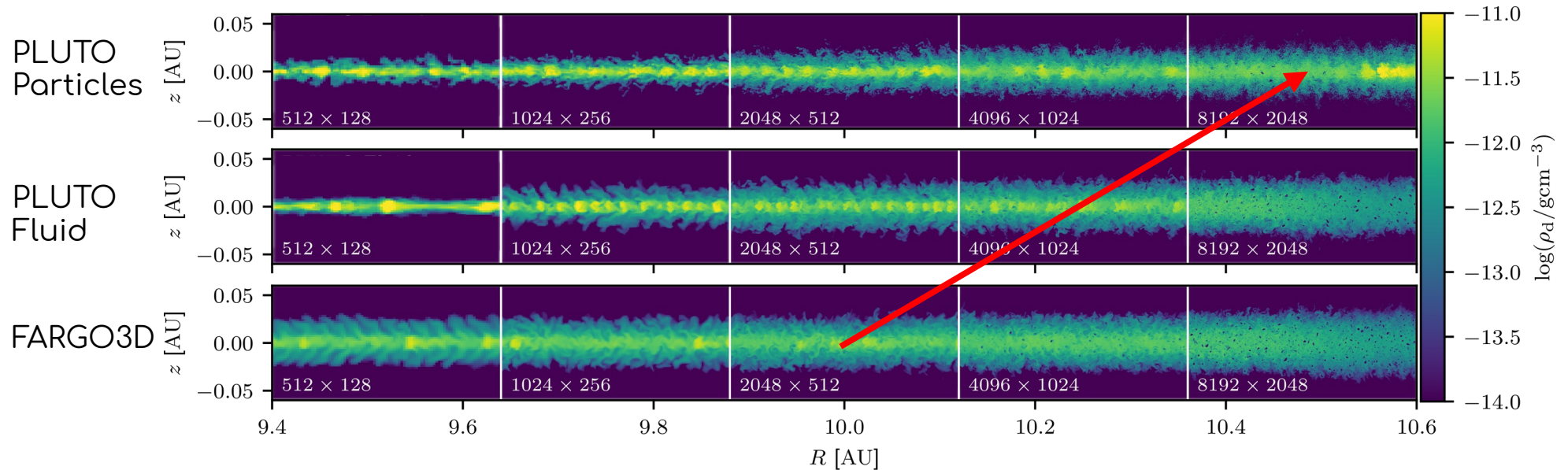
Fluid / Particles



- PLUTO Fluid also loses midplane
- Particles still thinner dust disk + broadened midplane component

see [Wang, Ormel et al. 2025](#) for better dust refill method using fixed flux of dust

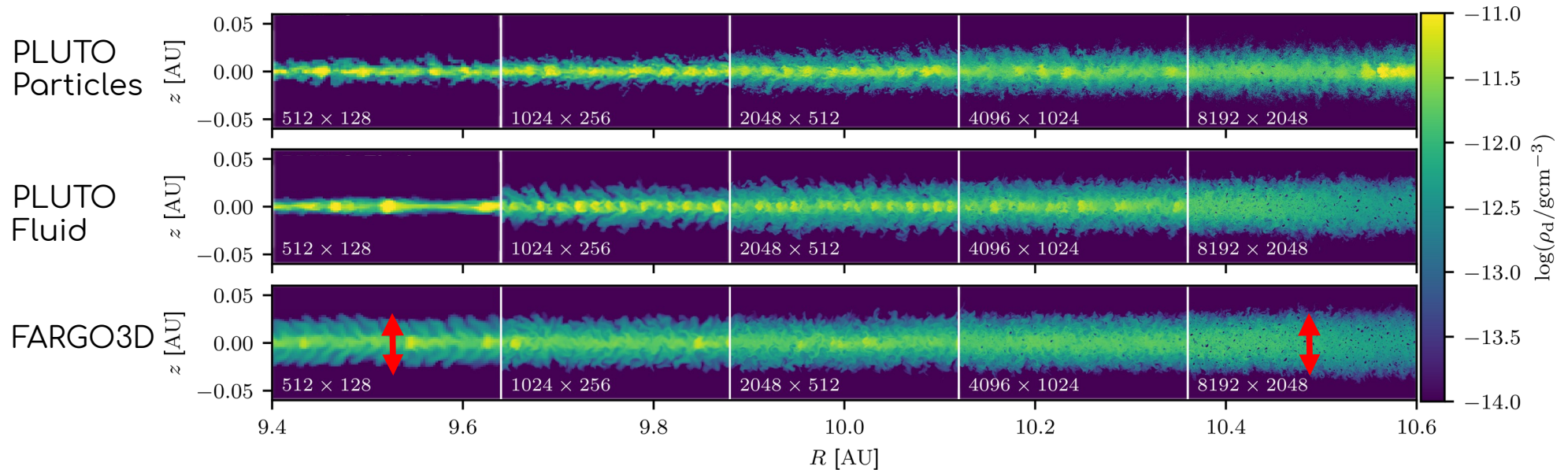
Fluid / Particles



- PLUTO Fluid also loses midplane
- Particles still thinner dust disk + broadened midplane component

see [Wang, Ormel et al. 2025](#) for better dust refill method using fixed flux of dust

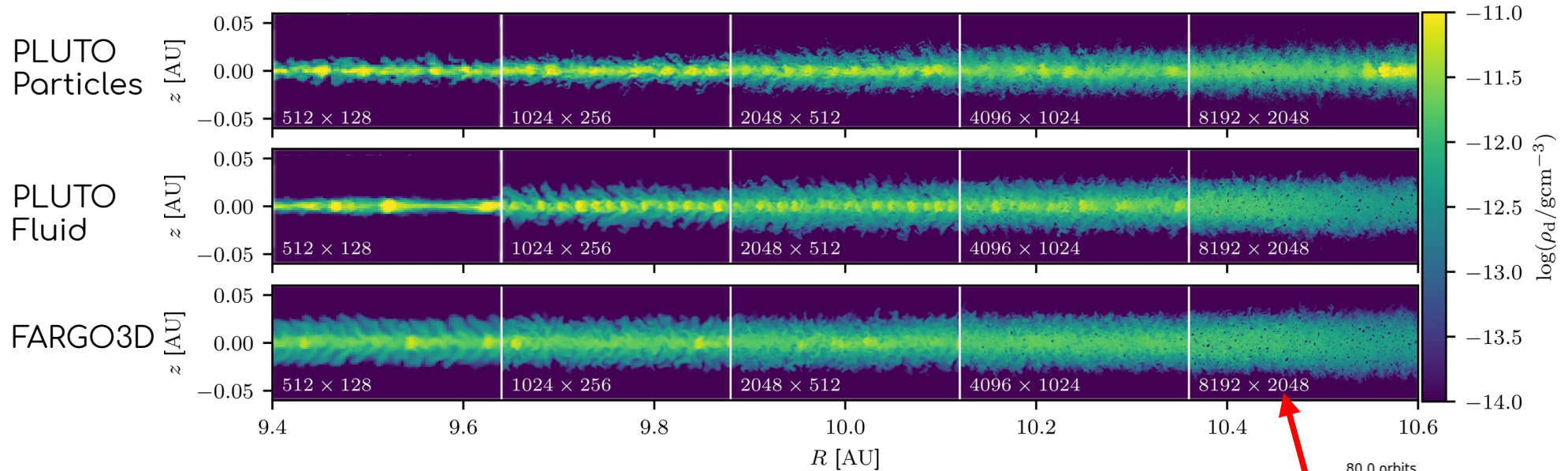
Fluid / Particles



- PLUTO Fluid also loses midplane
- Particles still thinner dust disk + broadened midplane component

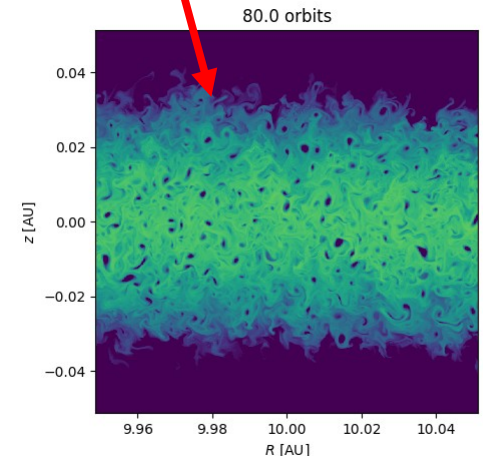
see [Wang, Ormel et al. 2025](#) for better dust refill method using fixed flux of dust

Fluid / Particles

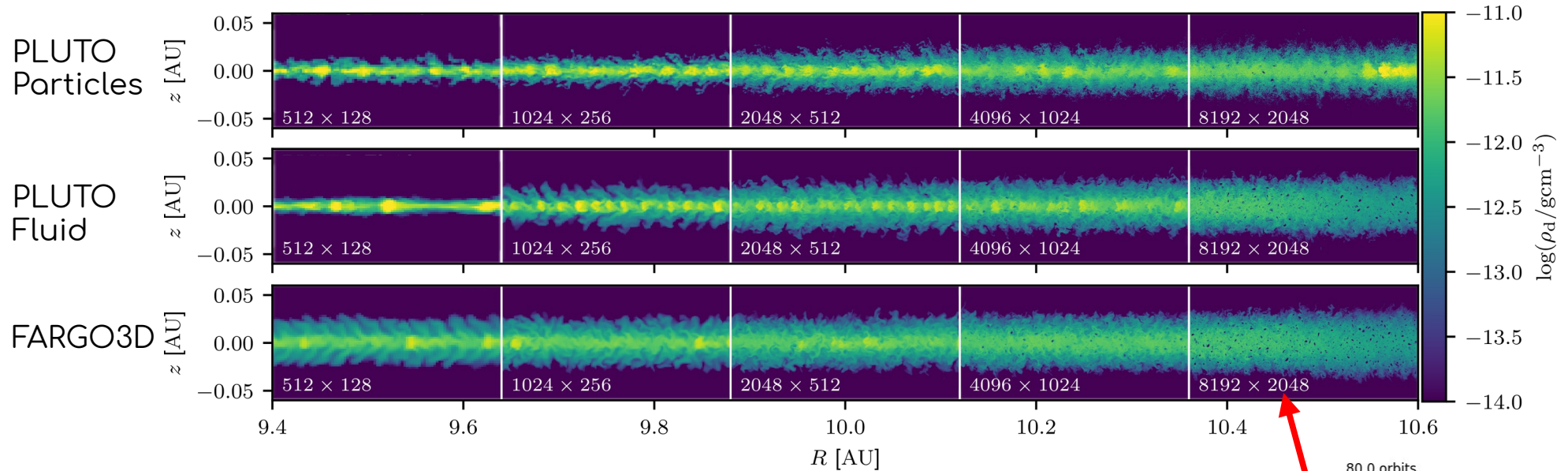


- PLUTO Fluid also loses midplane
- Particles still thinner dust disk + broadened midplane component

see [Wang, Ormel et al. 2025](#) for better dust refill method using fixed flux of dust



Fluid / Particles



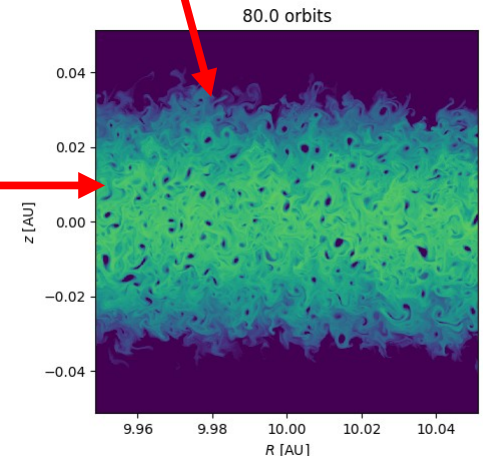
- PLUTO Fluid also loses midplane
- **Particles** still **thinner** dust **disk** + **broadened** midplane **component**

Squire+Hopkins 18:

$$k_x \eta r \sim 1/St$$

$$1/St = 100, k_x \sim 1/\Delta x$$

$$k_x \eta r \approx 360$$



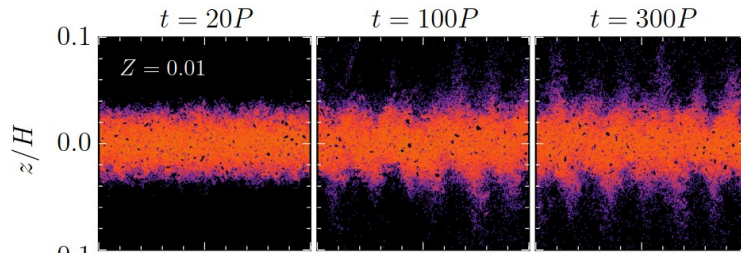
see [Wang, Ormel et al. 2025](#) for better dust refill method using fixed flux of dust

2 Discussion points

Particle simulations

FARGO3D vs. PLUTO

- [Yang+17](#): also see voids and no midplane for particles

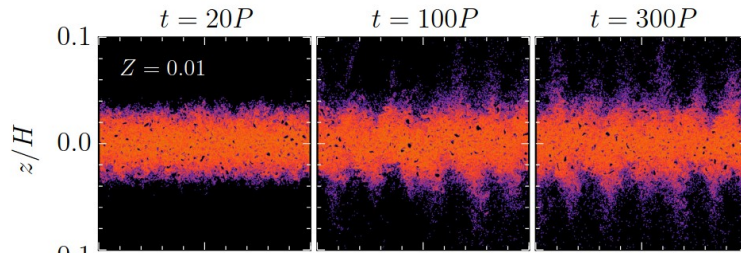


- Settling $\rightarrow$ high #particles/cell at midplane, but low numbers at higher z , pointed out by [Pinghui](#)
- My guess: either higher #particles or higher resolution

2 Discussion points

Particle simulations

- [Yang+17](#): also see voids and no midplane for particles



- Settling $\rightarrow$ high #particles/cell at midplane, but low numbers at higher z , pointed out by [Pinghui](#)
- My guess: either higher #particles or higher resolution

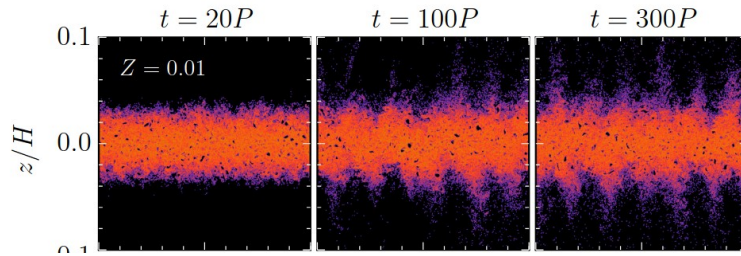
FARGO3D vs. PLUTO

- [Seligman & Laughlin 17](#): vortex dissipation higher for PLUTO wrt. ZEUS (velocity is staggered quantity)
- For low resolution, vorticity of FARGO3D may be higher wrt. PLUTO $\rightarrow$ larger H_d ?
- Should we use ZEUS-like codes like FARGO3D to study subsonic turbulence?

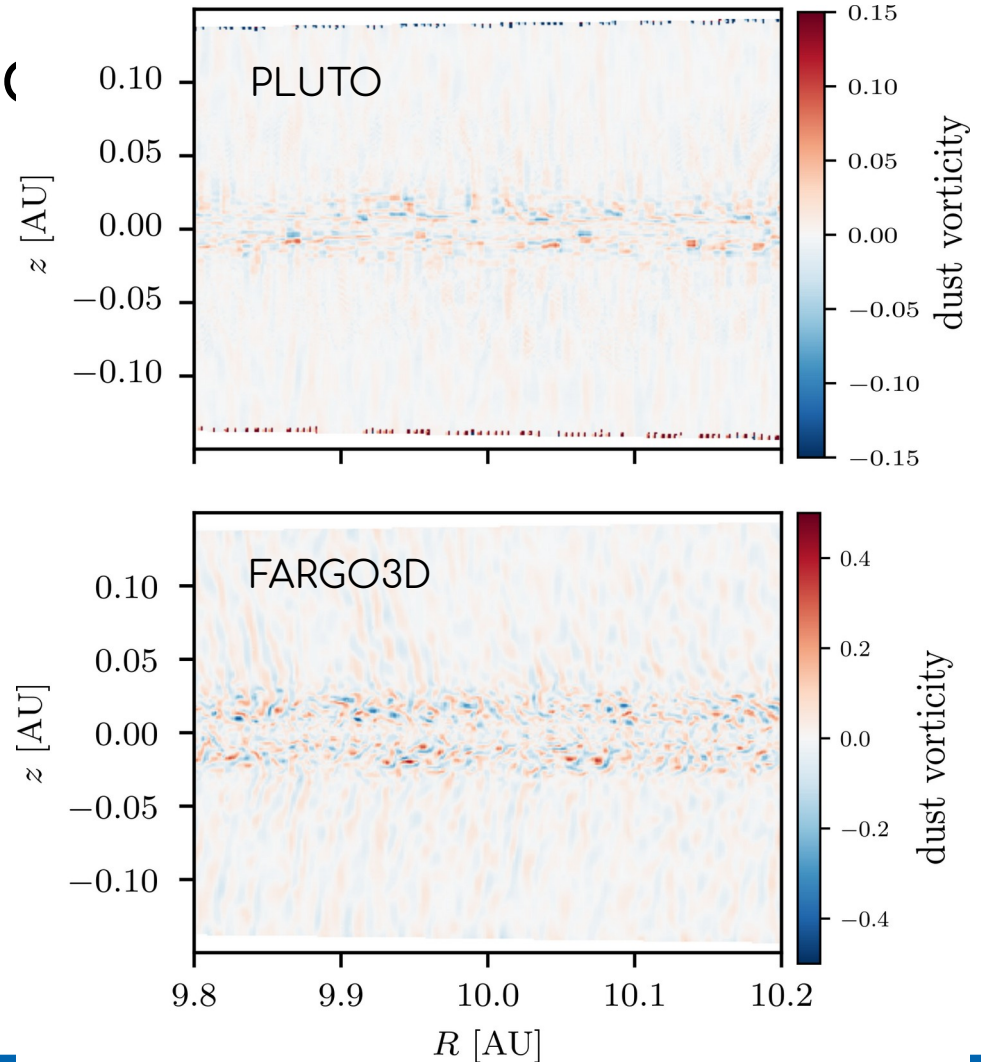
2 Discussi

Particle simulations

- [Yang+17](#): also see voids and no midplane for particles



- Settling → high #particles/cell at midplane, but low numbers at higher z , pointed out by [Pinghui](#)
- My guess: either higher #particles or higher resolution



Quick Note

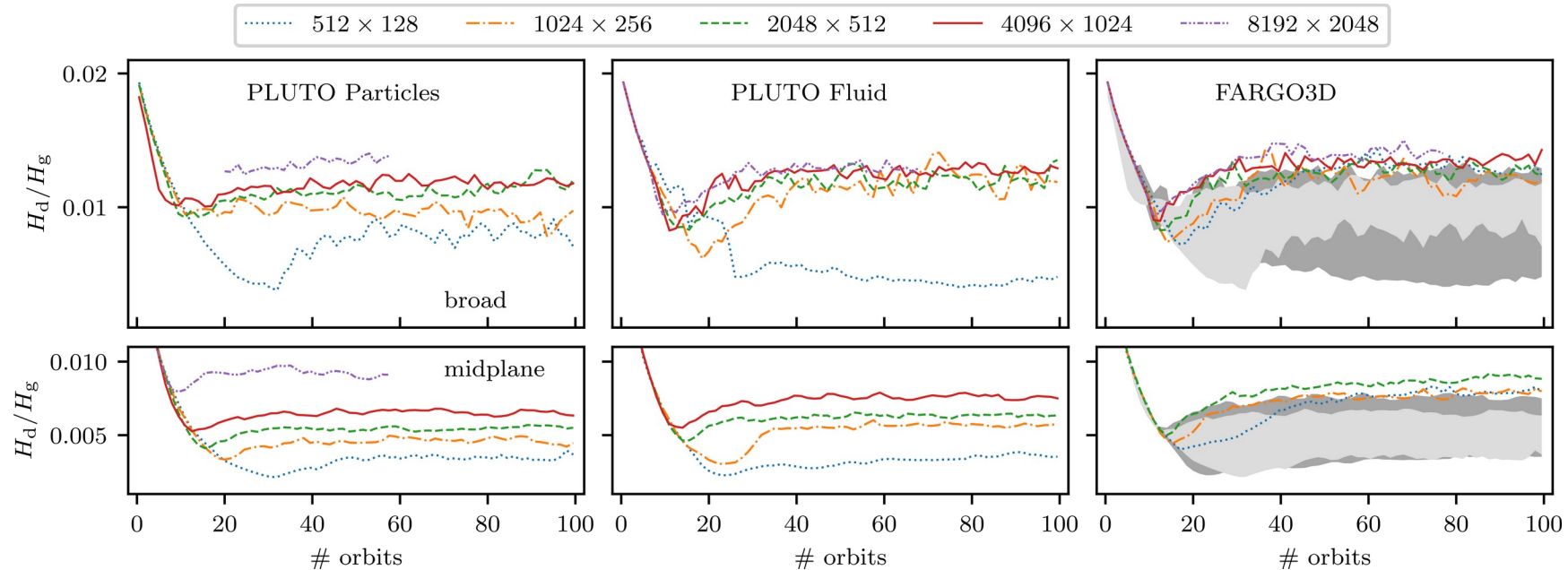
- As I'm in my final year → I look for PostDoc Positions for (late) next year
- Let me know about upcoming positions via email (ostertag@mpia.de) and/or slack

Conclusion

- Higher order **reconstruction**
more accurate **Riemann solver** } resolves **midplane component** (see also [Lim et al. 2025](#))
+ higher dust-to-gas ratio
- Different Approaches (for PLUTO) produce similar results
BUT: PLUTO Particles: thinner disk
- Convergence of broad component ✓ + able to resolve voids
No convergence of midplane component ✗ → **transient feature**
- Results from [Lim et al. 2025b](#): critical density for collapse is reached the fastest for least prominent midplane.
- [Daniel Carrera's](#) talk: need to consider also $St \sim 10^{-3}$ → even higher resolutions needed
- Q1: Which contribution to H_d/H_g is from numerical errors, which due to fluid approach (no crossing streams) → for future projects?
Q2: Zeus-like code better for subsonic simulations of turbulence?
- Q3: weakness of particle simulation as particles settle to midplane?

Backup slides

Sign of Convergence?

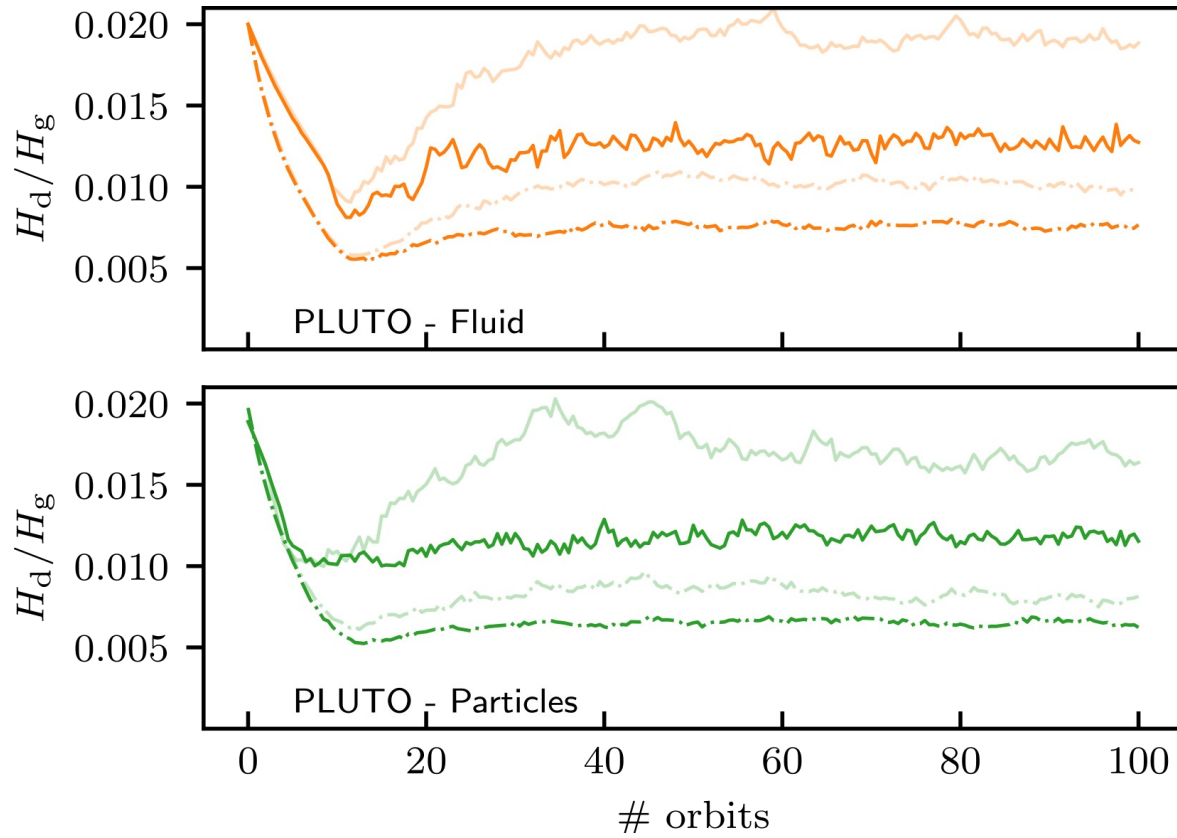


- Broad Density component shows convergence, but not midplane component
- See [Lim et al. 2025](#) for hints of vanishing midplane component in 2D

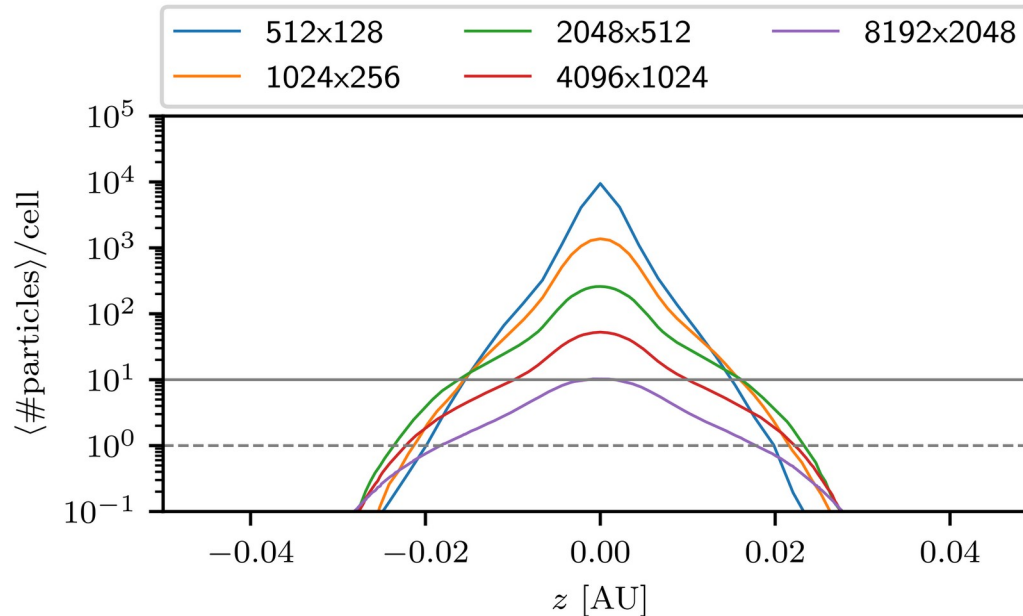
Average dtg ratio + dust density

- PLUTO Particles:
dtg: 1.34 max-dens: $8.9\text{e-}12 \text{ gcm}^{-3}$
- PLUTO Fluid:
dtg: 1.10 max-dens: $6.3\text{e-}12$
- FARGO3D:
dtg: 0.68 max-dens: $2.5\text{e-}12$
- PLUTO – PLM / PPM
dtg PLM: 0.78 dtg PPM: 1.06 max-dens PLM: $3.8\text{e-}12$ max-dens PPM: $6.5\text{e-}12$
- PLUTO – HLL / HLLC
dtg ROE: 1.08 max-dens ROE: $7.6\text{e-}12$

Influence of vertical domain size

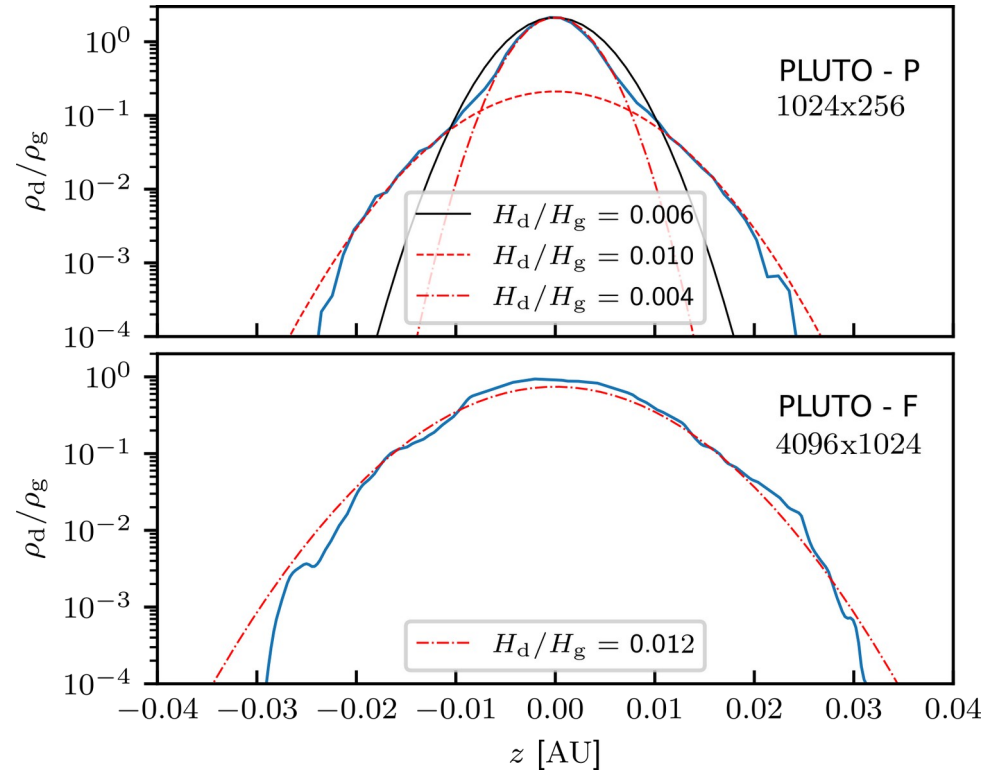


$\langle \# \text{Particles} / \text{cell} \rangle$

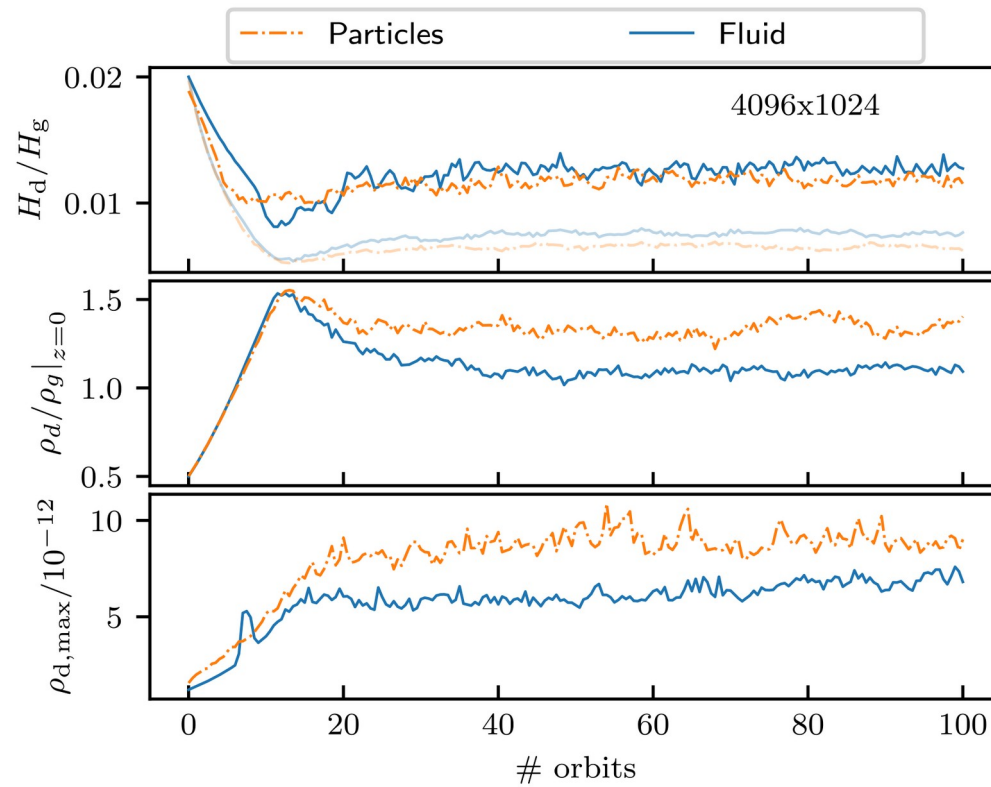


- Particle number = const = $1.04 \times 10^7 \rightarrow \# \text{particles/cell}$ decrease for resolution
- Q: does increase in initial $\langle \# \text{particles/cell} \rangle$ help for regions $z \sim 0.02 \text{ AU}$?
we are currently testing this with 4x more particles for 4096×1024

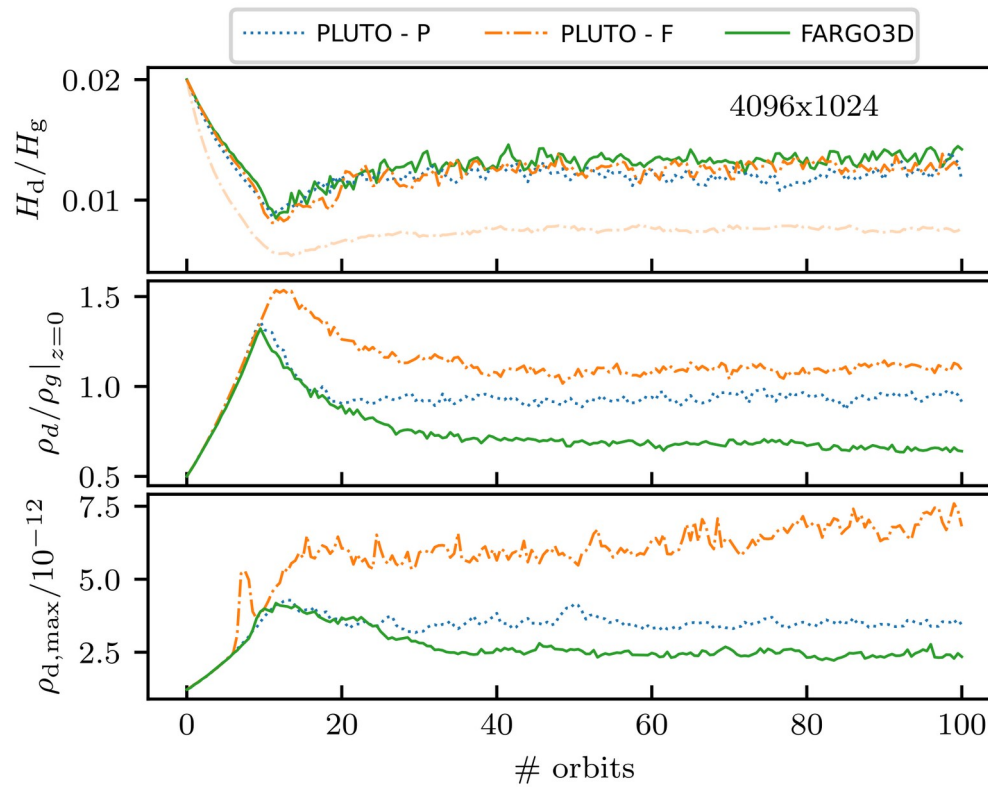
Determine Dust Scale Height



PLUTO - Fluid / Particles cont.



PLUTO - Fluid / FARGO3D cont.



Code Performance

Table 2. Performance comparison for the different dust modelling approaches and both hydrodynamics codes

Resolution	HW	#dts/orbit	#cells/step/s
PLUTO Fluid - PPM + RK3 + ROE			
512x64	AN	9791	$8.12 \cdot 10^6$
1024x128	AN	19603	$8.85 \cdot 10^6$
2048x256	AN	39272	$8.93 \cdot 10^6$
4096x512	AN	78679	$9.1 \cdot 10^6$
4096x512	VP	78376	$1.41 \cdot 10^7$
PLUTO Fluid - PLM + RK2 + HLL			
4096x512	AN	78278	$1.44 \cdot 10^7$

PLUTO Particles - PPM + RK2 + ROE			
512x64	AN	9794	33,578
1024x128	AN	19602	$1 \cdot 10^5$
2048x256	AN	39260	$4.93 \cdot 10^5$
4096x512	AN	78491	$2.19 \cdot 10^6$
4096x512	VP	78378	$4.01 \cdot 10^6$
PLUTO Particles - PLM + RK2 + HLL			
4096x512	AN	39151	$2.26 \cdot 10^6$
FARGO3D			
512x128	VA	5100	$8.7 \cdot 10^6$
1024x256	VA	10190	$9.72 \cdot 10^6$
2048x512	VA	20390	$1.01 \cdot 10^7$
4096x1024	VA	40770	$1.03 \cdot 10^7$
8192x2048	VA (1 GPU)	81540	$1.32 \cdot 10^7$

More about 1. Project

